

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

SUMMER SEMESTER, 2024-2025
FULL MARKS: 75

CSE 4621: Machine Learning

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all **3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Suppose you own a subscription-based company that delivers a curated book to each subscriber monthly based on individual reading preferences. You seek to classify subscribers who are likely to cancel or retain their subscriptions using two predictors: x_1 , the number of returned books, and x_2 , the reading-engagement score. The target variable is defined as $y = 1$ for cancellation and $y = 0$ for retention, as shown in Table 1. 5 + 5
(CO1)
(PO1)

Table 1: Training dataset for book-subscription cancellation classification for Question 1.a.

Subscriber	x_1 (Returned Books)	x_2 (Reading Engagement)	y
1	1	4.5	0
2	0	6.2	0
3	6	2.1	1
4	3	2.9	1

Assuming all parameters initialized to 0.5, answer the following questions based on Table 1:

- i. Compute the total loss of the dataset using binary cross-entropy loss. Compare this to the loss calculated using mean-squared error.
 - ii. Calculate the partial derivatives of the loss function with respect to each parameter. Using these gradients, carry out a single gradient descent update step with learning rate $\alpha = 0.1$.
- b) Explain how a neural network can approximate any continuous function defined on a bounded domain. Support your explanation with appropriate diagrams. 7
(CO1)
(PO1)
- c) Aviation investigators are analyzing a defective flight recorder recovered during a routine quality audit. Three companies manufacture all such flight recorders: AirCorp, BigSkies, and CharterAir, producing 80%, 15%, and 5% of the total supply, respectively. Historical quality-control records show that the defect rates for AirCorp, BigSkies, and CharterUS are 4%, 6%, and 9%, respectively. 8
(CO1)
(PO1)
- Given that the selected flight recorder is defective, determine which manufacturer it most likely came from.
2. a) An airline company is developing a system to predict whether a flight is likely to experience a significant delay. The neural network shown in Figure 1 is used for making such predictions. The network has three input features x_1 , x_2 , and x_3 . All neurons use the Sigmoid activation function. 8 + 4
(CO1)
(PO1)

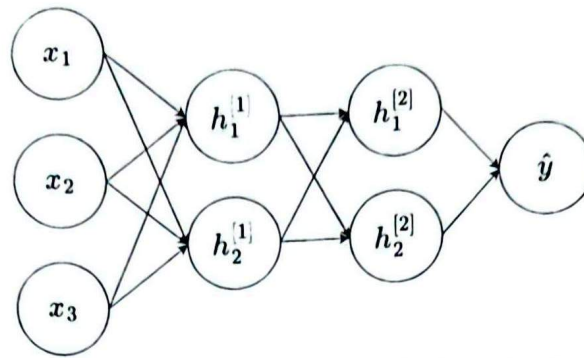


Figure 1: The neural network for Question 2.a

The layer-wise weights for the network in Figure 1 are:

$$W^{[1]} = \begin{bmatrix} 0.5 & -0.4 & 0.7 \\ -0.3 & 0.8 & 0.2 \end{bmatrix}, W^{[2]} = \begin{bmatrix} 0.6 & -0.5 \\ 0.4 & 0.7 \end{bmatrix}, W^{[3]} = [0.8, -0.6]$$

Assuming all the bias terms are set as 0.5, answer the following questions:

- Perform forward propagation on the network in Figure 1 on the input $x = [0.9, 0.7, -0.6]^T$ and predict the output $\hat{y}$.
 - Suppose the company decides to train the neural network using a combination of L1 and L2 regularization (Elastic Net regularization) rather than using only L1 or only L2 regularization. Explain how this choice would influence the learned network parameters.
- b) Using an appropriate mathematical proof, demonstrate that the cost function of a linear regression model is a convex function with respect to its parameters.
- c) Consider a color image of dimensions $512 \times 512 \times 3$ that is processed through the following sequence of layers:
- A convolutional layer with 8 filters, a kernel size of 3×3 , and a stride of 2.
 - A max-pooling layer with a kernel size of 2×2 and a stride of 2.
- Based on the above architecture, answer the following questions:
- Determine the total number of trainable parameters in the architecture. Clearly show your calculations.
 - Suppose the objective is to preserve the spatial dimensions of the input image after the convolution operation. Determine the amount of zero-padding required.
 - Determine the amount of zero-padding required to preserve the spatial dimensions of the input image.
3. a) An engine manufacturer developed a machine learning system that classifies the health status of a gearbox into one of four categories: Normal, Warning, Critical, and Failure. The model was evaluated on a test set of 300 gearbox observations and produced the confusion matrix from Table 2.

Table 2: Confusion matrix for gearbox status classification for Question 3.a.

		Predicted			
		Normal	Warning	Critical	Failure
Actual	Normal	80	12	3	0
	Warning	10	58	10	2
	Critical	2	12	42	14
	Failure	0	1	9	45

Answer the following questions based on Table 2:

- i. Calculate the Precision, Recall and F1 score for each class.
- ii. Examine the confusion matrix and identify the most common misclassification pattern. Explain what this suggests about the underlying data.

- b) Semantic segmentation is a computer vision task that classifies every pixel in an image into predefined categories. An autonomous vehicle uses a semantic segmentation model to identify pedestrians, vehicles, road markings, and obstacles.

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(CO3)
(PO3)

Design an appropriate evaluation metric for assessing the segmentation results. Provide the mathematical expression of the metric and explain how it measures the quality of the predicted segmentation compared with the ground-truth labels.

- c) Write short answers:

2 × 3
(CO1)
(PO1)

- i. What is the “curse of dimensionality”?
- ii. How is maximum log likelihood (MLE) used in mean squared error (MSE)?

Appendix for CSE 4621

A Probability

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

B Linear Regression

$$\hat{y} = h_{\theta}(x) = \theta^T x$$

$$J(\theta) = \frac{1}{2m} \|X\theta - y\|_2^2$$

C Gradient Descent

$$\theta \leftarrow \theta - \alpha \nabla_{\theta} J(\theta)$$

$$\theta \leftarrow \theta - \frac{\alpha}{m} X^T (\hat{y} - y)$$

D Regularization

$$J_{L2}(\theta) = J(\theta) + \frac{\lambda}{2} \|\theta\|_2^2$$

$$J_{L1}(\theta) = J(\theta) + \lambda \|\theta\|_1$$

E Logistic Regression

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\hat{y} = P(y = 1 | x; \theta) = \sigma(\theta^T x)$$

$$J(\theta) = -\frac{1}{m} \sum_{i=1}^m [y^{(i)} \log \hat{y}^{(i)} + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)})]$$

F Backpropagation

$$dZ^{[L]} = A^{[L]} - Y$$

$$dW^{[l]} = \frac{1}{m} dZ^{[l]} (A^{[l-1]})^T$$

$$db^{[l]} = \frac{1}{m} \sum_{i=1}^m dZ^{[l](i)}$$

$$dZ^{[l]} = (W^{[l+1]})^T dZ^{[l+1]} \odot g'^{[l]}(Z^{[l]})$$

G Regression Metrics

$$MAE = \frac{1}{m} \sum_{i=1}^m |y^{(i)} - \hat{y}^{(i)}|$$

$$MSE = \frac{1}{m} \sum_{i=1}^m (y^{(i)} - \hat{y}^{(i)})^2$$

$$RMSE = \sqrt{MSE}$$

$$R^2 = 1 - \frac{\sum_i (y^{(i)} - \hat{y}^{(i)})^2}{\sum_i (y^{(i)} - \bar{y})^2}$$

H Classification Metrics

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F_1 = 2 \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\text{TPR} = \frac{TP}{TP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN}$$